Nikky Amresh

Senior Software Engineer | Python, Node.js / TypeScript, Generative AI

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SUMMARY

Senior Software Engineer with 4+ years of experience building backend platforms in Python and Node.js. Currently at Educational Initiatives (Mindspark), working on the platform behind an adaptive learning product used by 500,000+ students, with a focus on microservices and AI-driven engineering automation. Outside work, built and shipped generative AI products, including a voice assistant, an AI assistant platform and a media generation pipeline, plus a mobile app live on both app stores.

EXPERIENCE

Senior Software Engineer, Platforms Engineering Dec 2023 - Present

Educational Initiatives (Mindspark), Bangalore

- Owned the backend for the Computational Thinking product, built in partnership with Amazon Feature Engineering, and shipped it to 100,000+ students.
- Built the Mindspark Control Panel to automate school code generation, demo tagging, CT-org mapping, and order and org lifecycle management.
- Built Mira Platform microservices for auth APIs, SMS notifications, approval workflows and event scheduling, on a common error-handling framework.
- Designed the Pre-Post Assessment System with impact tracking and permission handling.
- Built AI automation for the engineering team in Python and TypeScript: MCP servers for Jira triage, PR review and Confluence, and a debugging tool that analyses ELK logs, tickets and code to suggest root causes and fixes.

Python, PHP, TypeScript, Node.js, MongoDB, Postgres, Elasticsearch (ELK), AWS (EC2, S3), Docker, Microservices

PHP Developer Jun 2023 - Dec 2023

Wing Assistant, California (Remote)

- Built a real-time group and personal chat application for two web portals, with a scalable architecture and optimized database structures for low-latency delivery.
- Monitored and resolved Sentry issues and optimized backend performance.
- Implemented Pipedrive CRM integration and secure webhooks for communication between systems.

PHP, Python, JavaScript, MySQL, MongoDB, Docker, Sentry, Pipedrive

Full Stack Engineer Apr 2022 - May 2023

HackerEarth, Bangalore

- Built 40+ full-stack projects (Python/Django, Laravel, MERN, Rails) and 20+ DevOps scenarios for the assessment platform.
- Led knowledge-transfer sessions and assessed candidates on technical depth and problem solving.

Internships: Gramoday (Backend), Fotonicia (Software Engineer) Sep 2019 - Dec 2021

- REST APIs in Express and MongoDB, CI/CD on Bitbucket and AWS, Python ops automation; Magento plugins built while working directly with customers.

GENERATIVE AI PROJECTS

Kitten, AI Assistant Platform (Python, FastAPI, PostgreSQL/pgvector, MCP, Flutter, Docker)

- Email triage pipeline that combines rule-based filtering with LLM classification, using schema-validated model output, circuit breakers and shared rate limiting.
- Hosted MCP server that exposes memory, tasks and reminders to external AI agents, with push notifications through a Flutter app.

Canvas, Collaborative AI Workspace (TypeScript, Node.js, PostgreSQL)

- Infinite canvas where each card is a turn of an AI conversation, with branching, real-time presence, comments and sharing, plus an MCP endpoint for external agents.

Nova, Voice Assistant (Python, Pipecat, FastAPI, WebSockets, Whisper, TTS)

- Real-time speech-to-speech assistant with streaming audio and interruption handling.

Redline, AI Plan Review Tool (Python, open source)

- Browser-based review for plans and prototypes produced by AI coding agents: versioned documents, inline comments and suggested edits sent back to the agent.

Storyboard, Generative Media Pipeline (Node.js, ComfyUI, FLUX, FFmpeg)

- Turns a story prompt into scenes, consistent character images, captions, narration and a rendered video, with automated Instagram publishing.

OTHER PROJECTS

Twine, Mobile App (Flutter, Cloudflare Workers, D1, R2, libsodium): end-to-end encrypted app for couples, live on Google Play and the App Store.

LeetCode for VS Code (TypeScript): extension with 11,000+ downloads across the VS Code Marketplace and Open VSX; 40+ commands, study plans, AI solution explanations, cloud sync and a timed interview mode.

SKILLS

Languages: Python, TypeScript, JavaScript, PHP, Dart, C++

Backend & Data: FastAPI, Flask, Django, Node.js, Express, microservices, REST APIs, WebSockets, Postgres, MongoDB, MySQL, Elasticsearch

Generative AI: LLM APIs (OpenAI, Anthropic, Groq), AI agents and tool calling, MCP, embeddings and vector search (pgvector), LLM evaluation, speech-to-text and TTS, ComfyUI, FLUX

Cloud & Mobile: AWS (EC2, S3, Lambda), Cloudflare Workers, Docker, Linux, GitHub Actions, Flutter

Working style: mentoring and knowledge sharing, interviewing candidates, working directly with customers, end-to-end ownership

EDUCATION

B.Tech, Computer Science Engineering, Roorkee Institute of Technology, 2019 - 2023 (CGPA 8.8)

ACHIEVEMENTS

- **NECTF Organizer:** organized an international jeopardy-style cybersecurity CTF with 400+ participants from 25+ countries, online and offline, with a 2-person core team.
- **Smart India Hackathon 2022, Grand Finalist:** built Verifi, an offline OTP authentication solution for DRDO.
- **Chess:** regional-level player.